(i) $\int_{-2}^{1} f(x)dx$ (ii) $\int_{1}^{2} f(x)dx$

(iii) $\int_{2}^{3} f(x)dx$ (iv) $\int_{-2}^{1.5} f(x)dx$

(v) $\int_{1}^{3} f(x)dx$

Solution:

(i) $\int_{-2}^{1} f(x)dx = \int_{-2}^{1} x^2dx = \left[\frac{x^3}{3}\right]_{-2}^{1} = \frac{1}{3} - \left(\frac{-8}{3}\right) = 3$

(ii) $\int_{1}^{2} f(x)dx = \int_{1}^{2} xdx = \left[\frac{x^2}{2}\right]_{1}^{2} = \frac{4}{2} - \frac{1}{2} = \frac{3}{2}$

(iii) $\int_{2}^{3} f(x)dx = \int_{2}^{3} (x-4)dx = \left[\frac{x^2}{2} - 4x\right]_{2}^{3}$

$= \left(\frac{9}{2} - 12\right) - \left(\frac{4}{2} - 8\right)$

$= -\frac{15}{2} + 6 = \frac{-3}{2}$

(iv) $\int_{-2}^{1.5} f(x)dx = \int_{-2}^{1} f(x)dx + \int_{1}^{1.5} f(x)dx$

$= 3 + \int_{1}^{1.5} xdx$ using (i)

$= 3 + \left[\frac{x^2}{2}\right]_{1}^{1.5}$

$= 3 + \frac{2.25}{2} - \frac{1}{2} = 3 + \frac{1.25}{2} = 3.625$

(v) $\int_{1}^{3} f(x)dx = \int_{1}^{2} f(x)dx + \int_{2}^{3} f(x)dx$

$= \frac{3}{2} + \left(\frac{-3}{2}\right) = 0$ using (ii) and (iii)

Exercise 2.8

I. Using second fundamental theorem, evaluate the following:

1. $\int_{0}^{1} e^{2x}dx$ 2. $\int_{0}^{\frac{1}{4}} \sqrt{1-4x}\, dx$ 3. $\int_{1}^{2} \frac{xdx}{x^2+1}$

4. $\int_{0}^{3} \frac{e^x dx}{1+e^x}$ 5. $\int_{0}^{1} xe^{x^2} dx$ 6. $\int_{1}^{e} \frac{dx}{x(1+\log x)^3}$

7. $\int_{-1}^{1} \frac{2x+3}{x^2+3x+7}dx$ 8. $\int_{0}^{\frac{\pi}{2}} \sqrt{1+\cos x}\, dx$

9. $\int_{1}^{2} \frac{x-1}{x^2} dx$

II. Evaluate the following:

1. $\int_{1}^{4} f(x)\, dx$ where
$f(x) = \begin{cases} 4x+3, & 1 \le x \le 2 \\ 3x+5, & 2 < x \le 4 \end{cases}$

2. $\int_{0}^{2} f(x)\, dx$ where
$f(x) = \begin{cases} 3-2x-x^2, & x \le 1 \\ x^2+2x-3, & 1 < x \le 2 \end{cases}$

3. $\int_{-1}^{1} f(x)\, dx$ where $f(x) = \begin{cases} x, & x \ge 0 \\ -x, & x < 0 \end{cases}$

4. $f(x) = \begin{cases} cx, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$
Find 'c' if $\int_{0}^{1} f(x)dx = 2$

DO YOU KNOW?

$\int_{a}^{b} dx = b - a$

2.2.2 Properties of definite integrals

(i) $\int_{a}^{b} f(x)dx = \int_{a}^{b} f(t)dt$

(ii) $\int_{a}^{b} f(x)dx = -\int_{b}^{a} f(x)dx$

(iii) If $f(x)$ and $g(x)$ are continuous functions in $[a, b]$, then
$\int_{a}^{b} [f(x) \pm g(x)]dx = \int_{a}^{b} f(x)dx \pm \int_{a}^{b} g(x)dx$

(iv) If $f(x)$ is a continuous function in $[a, b]$ and a < c < b, then
$\int_{a}^{b} f(x)dx = \int_{a}^{c} f(x)dx + \int_{c}^{b} f(x)dx$

(v) $\int_{0}^{a} f(x)dx = \int_{0}^{a} f(a-x)dx$

Proof:

Take $a - x = t \Rightarrow dx = -dt$		
x	0	a
t	a	0

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